

ALERT BAY, BC, Canada

WMO: 741100

Lat: 50.582N

Lon: 126.915W

Elev: 195

StdP: 14.59

Time Zone: -8.00 (NAP)

Period: 86-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.2	(c) 28.2	(d) 14.0	(e) 11.3	(f) 27.6	(g) 19.5	(h) 14.8	(i) 29.5	(j) 58.8	(k) 43.5	(l) 57.1	(m) 42.5	(n) 9.6	(o) 140	(p) 0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.2	(c) 68.5	(d) 59.8	(e) 65.4	(f) 58.2	(g) 63.2	(h) 57.1	(i) 61.0	(j) 66.0	(k) 59.3	(l) 63.8	(m) 58.0	(n) 62.0	(o) 5.8	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.7	74.3	63.5	57.2	70.4	61.1	56.0	67.2	59.5	27.1	66.1	26.0	63.9	25.1	62.3	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
47.5	40.4	35.6	21.4	79.4	6.4	2.2	16.8	81.0	13.1	82.3	9.5	83.5	4.8	85.1
			19.1	66.7	6.6	2.5	14.3	68.5	10.4	70.0	6.7	71.4	1.9	73.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	39.4	40.1	43.1	46.2	50.6	53.5	57.0	57.9	53.8	48.9	43.1	39.7
	DBStd	7.62	4.50	6.53	3.78	3.69	3.54	2.71	2.50	2.69	3.34	4.27	3.78	4.74
	HDD50	1584	329	279	214	125	32	3	0	0	4	70	208	320
	HDD65	6268	794	697	678	563	445	344	247	222	336	499	658	785
	CDD50	793	0	1	1	12	52	109	218	244	119	36	1	0
	CDD65	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH74	23	0	0	0	0	1	3	4	6	9	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.5	17.2	14.1	11.3	8.8	8.0	6.6	6.7	5.6	5.7	10.7	15.9	15.4
Precipitation	PrecAvg	65.7	8.8	6.1	5.0	3.9	2.8	2.8	2.2	2.6	4.5	8.9	9.3	8.7
	PrecMax	86.4	18.0	10.3	9.0	6.9	5.7	6.8	5.9	6.1	9.0	18.1	26.6	15.9
	PrecMin	42.5	2.8	1.7	1.2	1.3	0.9	0.8	0.3	0.0	0.7	2.7	3.2	3.6
	PrecStd	10.0	3.7	2.5	2.0	1.4	1.3	1.4	1.5	1.6	2.2	3.7	4.7	3.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	55.1	56.4	64.3	69.1	67.5	72.0	73.3	73.4	64.8	53.1	49.8
		MCWB	45.6	48.4	47.8	51.2	58.1	59.2	61.8	62.5	57.5	57.6	51.1	46.4
	2%	DB	47.6	51.5	53.0	58.5	62.7	63.0	66.3	68.2	64.5	60.1	51.4	47.8
		MCWB	45.1	45.9	44.8	49.6	54.2	56.8	58.6	60.5	58.2	54.8	49.6	45.5
	5%	DB	46.3	49.4	50.7	54.9	59.5	60.6	64.0	65.3	61.3	57.0	49.7	46.8
		MCWB	44.0	45.1	45.2	48.0	53.1	54.7	57.6	58.8	56.2	53.2	47.9	44.7
	10%	DB	44.9	47.8	48.7	52.2	57.0	58.9	62.0	63.1	59.2	54.6	48.1	45.6
		MCWB	42.7	44.4	44.4	47.1	51.7	53.8	56.6	57.7	54.8	51.8	46.4	43.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	49.6	50.1	53.5	58.5	60.6	62.9	63.7	61.2	59.2	52.2	46.9
		MCDB	48.7	54.6	52.0	62.4	67.9	64.8	68.4	71.3	66.9	62.9	53.2	48.2
	2%	WB	45.6	47.2	47.5	50.7	55.8	57.7	59.9	61.2	58.9	56.0	50.0	46.0
		MCDB	47.1	50.1	50.4	56.3	61.2	62.3	65.0	66.4	62.7	58.7	51.0	47.5
	5%	WB	44.1	45.8	46.4	49.2	53.9	55.9	58.2	59.8	57.1	53.6	48.1	45.1
		MCDB	45.8	48.5	48.9	53.5	58.0	59.1	62.8	64.0	60.6	55.9	49.2	46.6
	10%	WB	43.0	44.7	45.1	47.7	52.4	54.7	57.0	58.4	55.6	52.1	46.7	44.0
		MCDB	44.8	47.2	47.7	51.8	56.6	57.7	61.0	62.1	58.5	54.4	47.9	45.5
Mean Daily Temperature Range	5% DB	MDBR	5.1	7.2	9.2	10.4	11.2	10.1	10.4	11.2	11.0	8.0	5.0	4.7
		MCDBR	5.8	9.0	13.2	17.2	17.0	13.7	14.5	15.8	15.1	12.7	5.7	4.8
		MCWBR	4.6	5.5	7.7	9.6	10.0	8.3	8.4	9.0	9.3	8.4	4.6	3.8
	5% WB	MCDBR	5.3	7.3	9.0	13.9	15.2	12.2	13.1	14.3	14.2	11.4	5.6	4.3
		MCWBR	4.5	5.1	6.3	8.5	9.5	8.1	8.5	8.8	9.1	8.0	4.7	3.8
Clear-Sky Solar Irradiance	taub	0.316	0.319	0.348	0.361	0.373	0.367	0.369	0.358	0.353	0.362	0.323	0.307	
	taud	2.423	2.497	2.411	2.399	2.397	2.450	2.480	2.527	2.524	2.390	2.442	2.403	
	Ebn at Noon	227	262	271	280	280	282	280	278	267	238	222	211	
	Edh at Noon	20	23	30	34	36	34	33	30	27	26	19	17	
All-Sky Solar Radiation	RadAvg	237	515	781	1201	1556	1629	1734	1508	1049	559	277	189	
	RadStd	32	87	108	97	173	178	215	130	99	93	49	29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page